

The Argument for Epistemic Competence

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How do we come to know? At what point does some encounter with information become knowledge for us? How can we be certain of what we know? When do we know that we are competent in addressing such questions about knowledge and knowing? These types of questions have been plaguing philosophers since the time of Plato, who defined knowledge as justified true beliefs (Plato, 1993). For teachers and educational researchers, philosophical discussions of epistemology and the nature of knowledge have taken the background to more pragmatic discussions of students' and teachers' beliefs about knowledge and knowing (i.e., epistemic beliefs; Hennessey, Murphy, & Kulikowich, 2013; Hofer & Pintrich, 1997). Yet, concerns about justifications and what qualifies as an appropriate justification for the myriad tasks in which students engage remains a question not relegated to philosophical inquiry. Rather, understanding beliefs about justifications for knowledge is important for today's learners and those who teach them (Maggioni, Fox, & Alexander, 2010). Such issues may be even more pressing for those who must navigate a world marked by an incessant flow of information and the concomitant demand to stay afloat within that informational deluge—a condition that seems to favor an expedient search for a plausible answer rather than one that is well justified (Alexander et al., 2011; List, Grossnickle, & Alexander, under review). In this chapter we briefly review research on epistemic beliefs, present an explanation of epistemic competence as it differs from traditional conceptualizations of epistemic beliefs, address the ways in which epistemic competence may be of particular importance for 21st century learners, and provide an explanation for how epistemic competence can be promoted in educational contexts.

The past four decades have brought a growing interest in describing the nature of epistemic beliefs (e.g., Baxter Magolda, 1992; Greene, Torney-Purta, & Azevedo, 2010; Perry, 1970). To this end, measures of epistemic beliefs have frequently been used to unearth and categorize individuals' beliefs about knowledge and knowing (Schommer, 1990; Schraw, Bendixen, & Dunkle, 2002; Wood & Kardash, 2002). Despite their differences, this catalogue of measures has been built around two underlying assumptions that require reflection. The first is that epistemic beliefs develop in a trajectory from least to most sophisticated (King & Kitchener, 2004; Perry, 1970). Within the epistemic beliefs literature, individuals are regarded as starting out naïve and developing in sophistication over time and as a result of academic experience (Greene et al., 2010; Hofer & Pintrich, 1997). For instance, those who are epistemically naïve

would judge that knowledge is rather simple in nature and see authorities as a good source of justification. In contrast, epistemically sophisticated individuals would appreciate the complexity of knowledge and require strong evidence to support any knowledge claim.

The second assumption is that the statements made in response to the questions asked reflect rather stable beliefs about knowledge and knowing that presumably guide individuals' actions in the real world (DeBacker, Crowson, Beesley, Thoma, & Hestevold, 2008). That is to say, if certain individuals respond that they believe knowledge is simple and dependent upon justifications by authority, one should feel confident that they would enact such beliefs rather consistently across a range of situations and contexts. Conversely, individuals whose responses suggest that they believe knowledge to be complex and requiring the careful weighing of evidence are assumed to operate on such beliefs in any condition. A lack of contextual consideration has been a major critique of epistemic belief questionnaire assessments (Leach, Millar, Ryder, & Séré, 2000; Mason, Ariasi, & Boldrin, 2011).

There is quantitative and qualitative evidence that beliefs about knowledge and knowing do undergo change over time as a consequence of cognitive and educational developments (Burr & Hofer, 2002; Conley, Pintrich, Vekiri, & Harrison, 2004; Kuhn, Cheney, & Weinstock, 2000). However, certain factors identified within the growing number of epistemic measures hold up to varying degrees under psychometric scrutiny (DeBacker et al., 2008), and such an approach to capturing and classifying beliefs about knowledge and knowing seems incomplete and misguided. Specifically, what these perspectives on knowledge and knowledge overlook is that beliefs about knowledge and knowing, especially in terms of what counts as viable evidence or justification, does not operate in a fixed or unconditional manner. In contrast, epistemic competence suggests that the form and quality of evidence required to justify perceptions, interpretations, or inferences as “knowledge” varies as a consequence of task and context.

Indeed, perceptions of epistemic beliefs as a stable set of beliefs have undergone scrutiny (Alexander & the Disciplined Reading and Learning Research Laboratory [DRLRL], 2012; Elby & Hammer, 2001; Hammer & Elby, 2003). One avenue of investigation has included questions about whether individuals' beliefs about knowledge and knowing remain consistent across domains such as mathematics and history (Buehl & Alexander, 2005). Recent research into domain-specific variations in beliefs has suggested that individuals' beliefs differ according to the domain in which their beliefs are enacted (Hofer, 2000; Muis, Bendixen, & Haerle, 2006;

Schommer-Aikens, Duell, & Hutter, 2005), although there is evidence that epistemic beliefs tend to be made up of both domain-general and domain-specific beliefs (Muis et al., 2006).

An additional avenue of inquiry into the stability of epistemic beliefs is that of contextual variation (Hammer & Elby, 2002; Louca, Elby, Hammer & Kagey, 2004; Mason et al., 2011). Much of the current research into the relation between context and epistemic beliefs has examined broad contexts such as culture (Hofer, 2008). To this extent, the focus has frequently been on stable differences across national or educational cultures (Bråten, Gil, Strømsø, & Vidal-Abarca, 2009; Haerle & Bendixen, 2008; Muis & Sinatra, 2008). More recent examinations into the stability (or adaptability) of beliefs across varied contexts have focused on identifying the specific contexts in which individuals enact their beliefs (Mason et al., 2011; Rosenberg, Hammer, & Phelan, 2006; Weinstock, 2009). Whether individuals' are cognizant of contextual factors influencing their beliefs and whether their epistemic beliefs reflect an understanding of the context as influencing the justifications required for knowledge has only begun to emerge within discussions epistemic beliefs (Elby & Hammer, 2001; Maggioni et al., 2010). Taking into consideration questions of contextual nuance that counter traditional notions of more and less sophisticated stable beliefs, epistemic competence has recently been forwarded as a means of re-envisioning epistemic beliefs (Alexander and the DRLRL; Maggioni et al., 2010).

What is Epistemic Competence?

Epistemic competence can be defined as the use of available contextual elements to determine the sources of evidence best suited for a given problem in an effort to provide an answer that is justified true belief. In this sense, epistemic competence focuses on the following two principles: (a) knowledge requires justification and (b) competent beliefs about knowledge and knowing require sensitivity to the context. Epistemic competence focuses on beliefs about knowledge and knowing rather than including beliefs about learning, harkening back to traditional notions of epistemology (Hofer & Pintrich, 1997).

Central to the idea of epistemic competence is the focus on knowledge as requiring justification (Alexander & the DRLRL; Maggioni et al., 2010), as its core principles suggest. Justifications are the standards that students hold to be sufficient to substantiate knowledge (Mason et al., 2011). The inclusion of justifications for knowledge in research on epistemic beliefs has been integrated as a component of epistemic beliefs research over the past several decades (e.g., Hofer & Pintrich, 1997; King & Kitchener, 1994). Research about epistemic

beliefs has identified various standards of justifications that students apply to knowledge. These have included beliefs that knowledge does not require justification, beliefs that knowledge is something to be verified by sources of authority (e.g., books, professors), and beliefs that evidence and opinions about knowledge from various perspectives should be verified through internal justifications and the weighing of evidence (Hofer & Pintrich, 1997; King & Kitchener, 2004). However, while these standards of justification have frequently been ranked in order of sophistication, epistemic competence differs in that it does not regard particular justifications as consistently superior (Alexander et al., 2011).

As discussed, perspectives on epistemic beliefs have commonly been based on the idea that individuals develop in sophistication from a complete trust in authority and use of authority as a justification for knowledge to complete distrust in authority and skepticism of any justifications for knowledge, and eventually to more moderate beliefs that evidence must be carefully considered and weighed (Hofer & Pintrich, 1997; Greene et al., 2010). In this sense, arguments for epistemic sophistication regard justifications via authority as naïve, and the weighing of evidence as the most desirable stage, albeit one that most individuals never achieve (Baxter Magolda, 2004). Epistemic competence sets aside the notion that more competent individuals unconditionally dismiss justifications via authority and provide more advanced types of justifications for knowledge regardless of the situation (Alexander et al., 2011). Instead, it regards the identification of appropriate justifications for the task at hand, be they justifications via authority or the careful weighing of evidence, as integral to competence (Maggioni et al., 2010).

The second main characteristic of epistemic competence is the understanding that the types of justification required necessarily depend on the context (Elby & Hammer, 2001; Louca et al., 2004; Maggioni et al., 2010). Epistemic competence moves away from the belief in privileging a certain type of justification under all conditions, to a determination of the best evidence for the current conditions (Elby & Hammer, 2001). That is, epistemic competence focuses on the centrality of justifications in determining knowledge and the belief that the type of justification necessarily depends on the task, context, and significance of the decision to be rendered (Alexander & the DRLRL, 2012; Maggioni et al., 2010). In this way, epistemic competence sets aside decontextualized notions of sophistication in favor of more complex and dynamic notions of epistemic development. It argues that those who consistently rely on some

external authority for justification are no more “sophisticated” in their epistemic beliefs than those who consistently dismiss an external authority as suitable justification. Rather, what develops in terms of epistemic competence is an understanding of what manner of justification is likely best suited for a certain problem posed within a particular context.

Let us take the writing of this chapter as a case in point. For instance, during the writing process we found it critical to weigh empirical and theoretical evidence from various perspectives and to provide reasoning based on the available evidence. However, for many situations, this type of justification is not advantageous, and the careful weighing of evidence and the belief that knowledge is complex and uncertain does not necessarily represent the best justification. For instance, when seeking to identify the year in which *Educational Psychologist* published a special issue on epistemic beliefs, we relied on authority, namely a search of the journal’s webpage. In this instance, the task at hand did not require a thorough weighing of evidence to ensure us that the knowledge was justified, rather, the identification of the authority appropriate for this situation lead us to reasonably accept a conclusion based on a type of justification often regarded as “unsophisticated.”

The expertise literature provides an additional case for examining the development of epistemic competence. For instance, characteristic of expertise is the ability to engage in reproducibly superior performance (Ericsson & Charness, 1994). While the high-level of consistent performance seems to align with notions of epistemic beliefs as complex and stable under traditional models of epistemic beliefs, the conditions under which reliably superior performance occurs necessarily differs across tasks and situations (Ericsson, Krampe, & Tesch-Römer, 1993). In this sense it requires what is defined as reproducibly superior performance to be adaptive to contexts, and the expert to be one who is capable of identifying justifications as appropriate to the context. As such, expertise requires a consistently reflective and flexible application of varying justifications, which aligns with the notion of epistemic competence. Moreover, for expertise, the nuanced application of appropriate justifications may extend beyond traditional domains to include the variety of situations and contexts in which experts typically engage, with less epistemic competence in areas outside the area of expertise.

The type of justifications best suited for a given problem may necessarily be related to the nature of knowledge required for the problem (Elby & Hammer, 2001; Greene et al., 2010). Such consideration includes the acknowledgment of whether problems are well-structured (i.e., a

correct answer can be provided with a high level of certainty) or ill-structured (i.e., there is no correct answer and answers may only be tentatively supported; Buehl & Alexander, 2005; Muis et al., 2006). The ability of individuals to differentiate problems according to their structure has been related to the type of justifications that they provide, and individuals who are unable to differentiate problems in this way typically provide justifications indicative of a belief that all problems are well-structured (King & Kitchener, 2004).

Considerable within-domain variation in the nature of problems exists, yet trends across domains may be present (Elby & Hammer, 2001). For instance, mathematics, due to the nature of knowledge within this domain as more slowly evolving at this point in the field's history, may more often require verification indicative of well-structured problems (Elby & Hammer, 2001). However, even within well-structured domains, there is considerable contextual variation in knowledge to which epistemically competent individuals would attend (Maggioni et al., 2010). Rather than regard epistemic beliefs as stable beliefs endorsed by individuals, some researchers have suggested that they are better described as resources activated within given contexts, and which cannot be identified without consideration of the context (Hammer & Elby, 2003; Louca et al., 2004). Epistemic competence builds on this idea, suggesting that there is a form of meta-knowledge held by competent individuals that guides their awareness of the context and the appropriateness of corresponding justifications (Scherr & Hammer, 2009). Given the focus on context, we now turn to the educational context of the 21st century to describe what epistemic competence may look like for learners in these contexts, and the challenges that face learners and educators in their quest for competence.

Epistemic Competence in the 21st Century

As the Internet has gained prominence in classrooms and students' academic lives, researchers have turned to examining epistemic beliefs as they manifest in digital contexts (e.g., Hartley & Bendixen, 2001; Kammerer & Gerjets, 2012;). Teachers and educational researchers alike have emphasized the need to examine how epistemic beliefs may impact student's experiences in online environments and have stressed the importance of understanding how learners may adjust their epistemic orientations in online contexts (e.g., Bråten Strømsø, & Samuelstuen, 2005; Hofer, 2004), evidencing their desire for students to develop epistemic competence. In examining how students' epistemic competence may develop in digital contexts two factors will be considered: first, the extent to which issues of justifications for knowledge

may be both more important and more challenging for students online; and second, which constellations of students' epistemic beliefs manifest and may be particularly critical in varying digital contexts.

Justifications in Online Contexts

The Internet context may make students' determining standards of justification for knowledge especially difficult, necessitating the examination of how students justify knowledge when accessing information online (Mason et al., 2011). This context may complicate determining standards of justifications for a number of reasons. First, there is a great deal of variability in information and sources available on the Internet (Metzger, 2007), and sources online can easily be made to look credible, even if they are not (Alexander & Tate, 1999). Further, when sources are presented on the Internet, source and author information is often unavailable and few objective standards of source quality exist (Alexander & Tate, 1999). While it is important to note that individual differences in specific skills or levels of expertise may have an impact on whether identifying the credibility of sources pose significant problems (Rouet, 2006), for students who are early in their academic development, there are certain general characteristics of this context that pose challenges for epistemic competence. First, given the potential difficulties associated with source and information evaluation, students' beliefs about justifications for knowledge are of particular consequence. Second, given the increased time and effort required to verify and evaluate information online (Graham & Metaxas, 2003), it is important that students' justifications match the appropriate standard of verification necessary across various contexts, a marker of epistemic competence. Drawing on these two characteristics of epistemic competence in turn, we consider how epistemic competence may manifest in digital contexts, and how these contexts may serve as barriers to the development of competence.

A number of criteria have been introduced to categorize the varying standards of justification that students may apply to substantiating knowledge on the Internet. These include personal standards for justifying knowledge such as verification based on prior knowledge (Bråten et al., 2005) or prior beliefs (Mason et al., 2011) as well as justifications based on external standards. These external standards reflect justification by consensus, including verification through corroboration from additional sources (Bråten et al., 2005), as well as justification by standards of inquiry, including the use of rational or reasoned logic (Bråten et al.,

2005) or through reliance on scientific evidence (Mason et al., 2011). Often times these varying standards of justification are classified according to their relative sophistication (e.g., Mason et al., 2011). Rather than ranking the relative sophistication of various standards for justification, considerations of epistemic competence would look to students' abilities to strategically adopt varying standards of justification based on considerations of the task and context. For example, justifications based on scientific evidence may be more important when researching risks associated with medications, whereas justification through corroboration may be more pertinent when selecting college classes. Indeed, domain and disciplinary standards for justifications may be salient for tasks within Internet contexts. For instance, for domains such as history, which emphasize the nature of texts and sources as authored (Britt & Aglinskias, 2002), the evaluation of the credibility of online sources may serve as a natural extension to disciplinary practices of evaluating print sources. Yet, new technologies have allowed for sources unique to the digital age (e.g., Twitter) and lack a standardization of where source features (e.g., author, publisher) can be found. Therefore, this requires the consideration of justifications in Internet contexts in tandem with the standards for domain-specific justifications in which Internet-based tasks are situated.

Providing appropriate justifications for knowledge in Internet contexts necessitates beliefs about the sources of knowledge that support appropriate justifications. These beliefs concern students' evaluations of where knowledge comes from and include students' judgments of both general source trustworthiness and specific source features such as author credibility (Bråten, Strømsø, & Britt, 2009). Students have been found to base these judgments on a variety of dimensions, specifically the popularity of the source, the authoritativeness of the source or author expertise, and the scientific nature of the source (Mason, Boldrin, & Ariasi, 2010). Studies examining students' source evaluations have reported that students make these kinds of epistemic evaluations fairly frequently, relative to other epistemic dimensions, and evidence flexibility in their abilities to adopt varying standards of source evaluation (List, Grossnickle, & Alexander, 2013; Mason et al., 2010). Although standards of source evaluation are often ranked in order of increasing sophistication (Mason et al., 2010), epistemic competence would be marked by students' abilities to adopt varied standards for source evaluation as demanded by the task and demonstrate purposefulness in their flexible adoption of these standards.

The 21st Century Context

Certain aspects of epistemic beliefs may be particularly pertinent in Internet contexts and critical for the understanding and development of epistemic competence. Specifically, students' beliefs about the nature of the Internet as a reliable source for simple, factual information, and about the standards of justification necessary for information on the Internet, have been implicated as two factors specific to online contexts (Bråten et al., 2005; Strømsø & Bråten, 2010). Positioned within a model of epistemic sophistication, standards for justification are commonly organized ranging from an epistemic stance that all information on the Internet is trustworthy and reliable to one where all information demands critical evaluation (Bråten et al., 2005; Kammerer, Bråten, Gerjets, & Strømsø, 2013). Although we share Bråten et al.'s (2005) contention that epistemic beliefs are specific to the context (i.e., the Internet), we would extend this notion to say that epistemic beliefs also manifest based on specifics of the online task.

In fact, the need for students to be sensitive to not only context but also task may be reflected in the mixed empirical findings regarding the relation between students "more sophisticated" epistemic beliefs and performance on tasks in digital contexts. For instance, while a study of Internet-specific epistemic beliefs found what were regarded as more sophisticated beliefs to relate to better attention to source features (Kammerer et al., 2013), another study found less sophisticated beliefs to relate to greater skill in identifying relevant information online (Bråten et al., 2005). The differential findings of the role of Internet-specific epistemic beliefs and information use may reflect the need to consider students' epistemic competence. Since epistemic competence considers the adoption of varying standards for justification and source evaluation to most appropriately meet the needs of the task and the Internet context, when locating relevant information online (as in Bråten et al., 2005), students may be advantaged in treating the Internet as an authoritative source for simple, factual information. However, when conducting research about an issue of personal, medical relevance (as in Kammerer et al., 2013), it may be more important for students to adopt an epistemic orientation wherein information requires greater corroboration and justification.

These findings suggest the need to consider students' epistemic competence or their capacity to strategically adjust their beliefs, adopting varying standards of justification and scrutiny of sources, based on considerations of the task and context. Beliefs reflective of epistemic competence would consider which aspects of the online context would demand which standards of justification, and apply these accordingly. For example, in researching a paper or

health information, students might use their knowledge of search engine source organization based on relevance, rather than quality, to apply a high level of scrutiny in selecting reliable sources and scrutinizing the scientific nature of information. Conversely, in researching the next day's forecast, students may use their knowledge of search engine source organization based on relevance, to select the first available source and verify the information relative to their personal expectations for the weather, a less sophisticated, yet appropriate, standard for justification given the task. However, students may not be able to make such nuanced contextualized distinctions between the types of justifications appropriate for varying question types (List et al., 2013). As such, there is a need to consider how students' development of epistemic competence can be supported.

Promoting Epistemic Competence

With an understanding of what epistemic competence is and why it matters, particularly in the lives and learning of 21st Century students, we now consider how this essential ability can be promoted within the educational context. That is, what can educators do to foster epistemic competence among students in learning environments? Our response is to suggest the following four important attributes of any learning environment that should help develop students' understanding of epistemic competence and arm them with the knowledge and strategies required to achieve this laudable end:

- Provide adequate time and sufficient rewards for reflection and justification;
- Call for the substantiation of knowledge claims within classroom discourse;
- Examine the forms of evidence valued within different domains and disciplines;
- Create situations where various modes of evidence are jointly weighed and discussed.

As these four attributes are described in turn, it is important to recognize that the dynamic nature of epistemic competence suggests that it will not follow the linear, stepwise progression from naïve to sophisticated, as represented in traditional models of epistemic beliefs (King & Kitchener, 2004; Perry, 1970). Rather, these attributes are aimed at the development of students' ability to identify and provide appropriate justifications across contexts and situations.

Adequate Time and Sufficient Rewards

In her efforts to explain why students did not behave very strategically when completing academic tasks, Garner (1990) put much of the onus on classroom cultures where quick and

automatic answers were valued over more effortful and timely responses. We can apply this same concern to epistemic competence. Specifically, if what is to be valued are claims or positions that are evidenced-based in nature and where “better” or “best” evidence is sought over any seemingly passable answer, then the time required to achieve such responses must be part and parcel of learning environments. In effect, how can we hold students accountable for superficial and quick answers or expedient treatment of information when such actions are commonplace occurrences within schools and classrooms (Alexander & the DRLRL, 2012)? What is valued and rewarded in classrooms gets practiced and, regrettably, habituated. Quick, off-the-cuff responses are too frequently adequate to earn academic credit in many classroom cultures where “covering the content” is routine (Alexander, Murphy, & Woods, 1996; Durkin, 1978-1979). As a result, “good” students need not be epistemically competent under those less than rigorous academic conditions.

Thus, in order for learning environments to promote epistemic competence, there must be a conscious effort by educators to allow time for deeper reflection and evidence-based reasoning. Moreover, reasoned and justified responses should be consistently rewarded over those responses that are merely passable or rapidly generated. Such classroom cultures may be particularly necessary for today’s learners, as educators looking to foster epistemic competence are at least in part tasked with counteracting the epistemic culture of the Internet that seemingly rewards speed and relevance over deliberation and quality evidence (Fast & Campbell, 2004; Metzger, Flanagin, & Zwarun, 2003).

Substantiation of Claims

Question asking and question answering are mainstays of classroom discourse. There is certainly much evidence that students are often called upon to demonstrate their understanding of academic content by responding to the oral or written questions posed by their teachers—their educational authorities (Murphy, Wilkinson, Soter, Hennessey, & Alexander, 2009). Even in those instances when students are urged to take a greater role in classroom discourse, critical analytic thinking is not frequently observed (Murphy et al., 2009; Newmann, 2012). In essence, more talk in classrooms does not ensure that more quality talk, from an epistemic standpoint, is occurring. If the goal in classrooms is to foster epistemic competence, then students must come to understand that simply forwarding an opinion or even offering an adequate response is not sufficient. Rather, what they must recognize is that better answers are those that come with

support or justification. “I think,” by itself, does not stand as evidence. “I think because” must become routine in these learning environments, where the “because” entails appropriate support for whatever judgment or position is espoused. Achieving this goal of evidence-based thinking and critical analytic talk within classrooms also requires that answers offered by teachers and by peers not be accepted at face value or unconditionally. Students and teachers should populate a learning environment where it is acceptable to question the responses or evidence offered by others.

Granted it is not easy to forge a learning environment where claims are routinely substantiated and where those substantiations are thoughtfully and critically weighed. But, as we have argued, students are quite adaptable and they can come to understand that empty statements or unsupported opinions are not valued academic qualities. This is even more likely when students are provided with models of evidentiary-based thinking. When students bear witness to epistemic competence in the form of their teachers’ words and action; when the nature and attributes of epistemic competence are discussed and illustrated within the classroom context; and when students are encouraged to try their hand at such evidence-based thinking under the tutelage of their teachers, then epistemic competence is more likely to emerge and develop over time.

Domain and Disciplinary Standards

One of the salient attributes of epistemic competence is the understanding that there are standards of evidence that characterize all academic domains and disciplines (Elby & Hammer, 2001; Maggioni et al., 2010). For instance, when confronted with a problem in such scientific domains as chemistry or biology, there is typically a need to bring physical evidence to bear in solution. By comparison, in history, where there are incalculable interpretations for a historical event or situation, primary and secondary sources must be systematically analyzed to ascertain what interpretation is most credible. Further, when the problem is drawn from the field of mathematics, there may be the search for a mathematical “proof,” which entails deductive reasoning from givens. Simply stated, what distinguishes one domain from another or one discipline for another includes the nature of the problems confronted and the evidence that is deemed acceptable in the solution of those problems.

In learning environments that promote epistemic competence, these features of academic domains and disciplines need to become part of the classroom discourse. Moreover, the

structuredness of various domains and disciplines must be openly discussed so that students can begin to see the conditional nature of evidence (Alexander, 1998; Frederiksen, 1984). By structuredness, we are referring to the degree to which the problems that typify a domain have commonly agreed upon answers or not or more algorithmic pathways to solution or not. Of course, no domain or discipline is invariantly well-structured or ill-structured in form (Elby & Hammer, 2001). Within any domain or discipline, there is certainly a range of tasks that are more or less structured or more open or closed in nature. Thus, rather than simply teaching the *content* of biology, history, or mathematics, educators should make the nature or ontology of those domains and disciplines part of the school curriculum. Teachers should not simply assign tasks or problems but devote time and attention to the character of those tasks and problems so that students are better prepared to seek the form and depth of evidence suggested by the context. Modeling discipline-specific approaches to completing tasks, evidencing epistemic competence, requires teachers to demonstrate how sources of evidence are selected and evaluated in different domains and how different domains define justified true beliefs. Students who come to understand the nature of domains and disciplines or the characteristics of various tasks or problems associated with those academic areas, and the sources and evidence valued within those communities of practice, should, thus, be better prepared to demonstrate epistemic competence.

Comparison of Evidentiary Alternatives

If there is one lesson learned from the extensive literature on conceptual change, it is that students may be better equipped to process ideas more deeply when presented with contrasting cases or conflicting evidence (Guzzetti, Snyder, & Glass, 1992; Hynd, Alvermann, & Qian, 1997; Palmer, 2003). If we apply that lesson to the area of epistemic competence, we could argue that students may come to appreciate that different conditions call for different forms and levels of justification when they have the opportunity to compare and contrast problem sets. In effect, educators might present students with a set of tasks that they believe represent distinct epistemic cases (e.g., more open or closed in form) and use those contrasting cases as a mechanism for teaching about evidentiary thinking and suitable justification. Alexander (2006) described this as problem profiling and recommended equipping students with the language of problem types (e.g., well-structured v. ill-structured; linguistic v. non-linguistic; schooled v. non-schooled) so that they are better able to classify and characterize the tasks at hand. To build

further epistemic competence, students should be encouraged to contrast problem-solving approaches characteristic of varying domains as they ponder questions of true value to them and the broader society (e.g., climate change, childhood obesity, or literacy). Helping students understand the differences in the evidence, approach, and justifications these varying disciplines adopt serves to engage students in conversations surrounding what counts as an answer under what conditions.

Such comparative analyses within classroom contexts should also embrace the value or consequences associated with the problem or task being considered. For example, is this a brainstorming or exploratory situation where students should feel comfortable in generating a large number of potential responses without significant academic cost and without careful justification? Or, is this a context where the demand to generate the “best” response is high and, thus, the demand for justification is great? Students should also be helped to explore their own personal motivations for engaging in problem solving and the individual differences that they bring to the task. For instance, is this an area or problem in a domain that is personally interesting to them? Do they self identify as particularly good in biology, history, or mathematics. Does their prior knowledge and past experiences with a topic or task afford them greater awareness of the mechanisms or avenues for justification?

What we have been trying to establish in this final discussion is that epistemic competence is a valued end in students’ development. But it is an end that should not be left to chance or which cannot be taken for granted even among those who are supposedly well educated. Competent students are not necessarily competent epistemically. That is because learning environments that promote the development of epistemic competence are seemingly markedly different from mainstream schools and classrooms and the online environments that seem to dominate students’ lives. These are learning environments where the reflection and justification are valued in terms of time and rewards allocated to them and where claims must be routinely accompanied by evidentiary support that is well matched to the domain and task. These are learning environments where evidence is weighed and critiqued, and where discussions about the nature of evidence common to domains and disciplinary communities are part of the classroom discourse. In sum, if we agree that epistemic competence is a laudable goal for the educated mind, then it must become an explicit and valued aspect of the school experience for all learners in all academic domains.

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